# Jonathan Liu

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## EDUCATION

**University of Washington** **Seattle, WA**
Bachelor of Science in Electrical and Computer Engineering | GPA: 3.93 *Sep 2024 – June 2027 (expected)*
Master of Science in Electrical and Computer Engineering (Combined BS/MS, Analog and Mixed-Signal) *Sep 2027 – June 2028*

- **Honors:** Frey Endowed EE Scholarship (2026–27), IEEE HKN Honor Society (Top 8% ECE), Tau Beta Pi Engineering Honor Society (Top 12.5%), Annual Dean's List (2024–2026).
- **Relevant Coursework: Analog IC Design (EE 435)**, **VLSI I (EE 476)**, Computer Architecture (EE 469), all in progress.

## ANALOG IC DESIGN

**Analog IC Tapeout, UW Analog Tapeout Sequence (EE 435/436/437)** *Sep 2026 – Summer 2027*

- Designing an **analog IC for industrial-foundry fabrication** in a 3-quarter sequence sponsored by **Apple, AMD, and TSMC**.

**Amplifier and Optical-Receiver Design in 45nm CMOS (EE 332)** *Spring 2026*

- Designed an **operational transconductance amplifier (OTA)** in **Cadence Virtuoso**, hand-sizing every transistor with **$\mathbf{gm/I_D}$** analysis; diagnosed a first design that missed spec and re-optimized it in **ADE Assembler** until it passed.
- Reused the amplifier as a **transimpedance amplifier** for a high-speed **optical receiver**, delivering **6x** the bandwidth of a resistor-only front end and keeping data readable where that design failed.
- Laid out a **CMOS inverter** in **FreePDK45** that passed **Calibre DRC and LVS** sign-off with zero errors.

**Semiconductor Engineering Practicum** **July – Aug 2026**
*Hiroshima University* *Hiroshima, Japan*

- Selected for a U.S.–Japan semiconductor consortium with **Micron Technology** and **Tokyo Electron**; hand-drew a **CMOS inverter layout** in **Siemens L-Edit**, fabricated it in a cleanroom, and confirmed working transistors on a **Keysight B1500A** wafer prober.

## RELEVANT EXPERIENCE

**Hardware Engineering Intern** **June – Sep 2026**
*Trellis Robotics* *Mountain View, CA*

- **Designed a BMS and motor driver board from scratch** for a pipe-inspection robot after diagnosing why the previous design destroyed itself at power-up; the new board worked **the day it arrived with no hardware rework**.
- Engineered **battery protection and precision current-sensing circuits** for the robot's lithium battery, then fixed shutdown and power-switch behavior by reprogramming a battery-monitor chip's built-in settings, **avoiding a costly board respin**.
- Architected a **Power-over-Ethernet robot board** that powers two motors and a motion sensor over one network cable, and caught a chip timing flaw that corrupted network data **before fabrication**.
- Developed a **novel 3D positioning system** that maps a robot's path inside metal pipe, **where GPS and compasses fail**, using only a free-floating motion sensor and tether length; achieved **centimeter-level accuracy**.
- Built the robot's **entire control software from scratch**: motor, battery, and sensor firmware, a Wi-Fi bridge, and a live-video dashboard.

**Hardware Lead** **Oct 2024 – Present**
*Husky Robotics Drone Subteam* *Seattle, WA*

- Led a 20+ person EE team and managed three custom PCBs in parallel: a **flight controller**, an electronic speed controller, and a power distribution board, including design reviews and the power and signal interfaces between them.
- Designed a 4-layer **STM32H743** flight controller in **Altium**: schematic, layout, 5 V/3.3 V regulation, current sensing, and the connector pinout for motors, radios, and sensors.
- Led **system integration** for a University Rover Challenge quadrotor: 6S power, 50 A ESCs, Pixhawk 6C, Jetson Orin Nano, **MAVLink**.

## PROJECTS

- **AC-DC Boost Converter (June 2026):** designed, simulated in **LTspice**, and **breadboarded** a two-stage converter delivering an adjustable **+10–21 VDC** rail at **0.4%** load regulation under **100 mV** ripple.
- **Three-Band Active Equalizer (March 2026):** built an **LM348** op-amp equalizer; hand-derived transfer functions, validated in **LTspice**.
- **FPGARhythm (March 2026):** developed a hardware rhythm game in **SystemVerilog** on an Intel **Cyclone V** FPGA with **VGA** output.
- **Line-Following Car (Feb 2026):** built a **PID**-tuned Arduino car; **all-time fastest record** in the UW competition.

## SKILLS

**IC Design & Test:** Cadence Virtuoso (schematic, layout, PEX), Calibre DRC/LVS, Spectre, ADE Explorer/Assembler, GPDK045/FreePDK45, Siemens L-Edit, LTspice/SPICE, $gm/I_D$ sizing, cleanroom CMOS fabrication, B1500A I-V wafer probing.

**PCB & Embedded:** Altium Designer, KiCad, multi-layer layout, multi-rail PDN, power sequencing, IPC-2221/DFM, STM32 (CubeIDE/HAL), ESP-IDF, Arduino, UART/$I^2C$/SPI/CAN, battery management, motor drivers, buck converters, LDOs.

**Digital & Software:** SystemVerilog, FPGA (Quartus, ModelSim), C/C++, Python (NumPy/SciPy), MATLAB, Linux/Bash, Git.

**Leadership: President**, IEEE UW Student Chapter; **Events Officer**, Hong Kong Student Association at UW.

**Languages:** English, Mandarin, Cantonese, Japanese.